CHAPTER 7

ROTATION OF RIGID BODIES

Key idea: torque, moment of inertia and angular momentum play the roles of force, mass and momentum.

1 ROTATIONAL KINEMATICS

$$\omega = \omega_0 + \alpha t$$

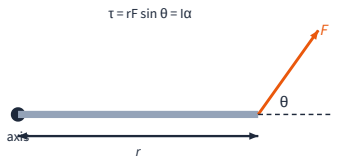
$$\theta = \omega_0 t + \tfrac{1}{2}\alpha t^2$$

$$\omega^2 = \omega_0^2 + 2\alpha\theta$$

Link to the rim:

$s = r\theta$ $v = r\omega$ $a = r\alpha$

2 TORQUE



$\tau = rF\sin\theta$ $\tau = I\alpha$

Unit N m; only the perpendicular component turns the body.

3 MOMENT OF INERTIA

$$I = \sum mr^2$$

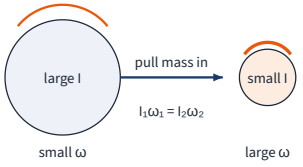
Body (axis)	I
Ring, hoop	MR^2
Disc, cylinder	$\frac{1}{2}MR^2$
Solid sphere	$\frac{2}{5}MR^2$
Rod, centre	$\frac{1}{12}ML^2$
Rod, end	$\frac{1}{3}ML^2$

Depends on mass, its distribution and the axis.

4 LINEAR ↔ ROTATIONAL

Linear	Rotational
s	θ
v, a	ω, α
m	I
$F = ma$	$\tau = I\alpha$
$p = mv$	$L = I\omega$
$\frac{1}{2}mv^2$	$\frac{1}{2}I\omega^2$
$W = Fs, P = Fv$	$W = \tau\theta, P = \tau\omega$

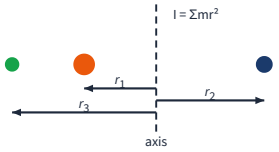
5 ANGULAR MOMENTUM



$L = I\omega$ $\tau = \dfrac{dL}{dt}$

Conservation: no external torque, $I_1\omega_1 = I_2\omega_2$ (skater pulls arms in: I ↓, ω ↑).

6 ROTATIONAL KE & ROLLING



$$E_k = \tfrac{1}{2}mv^2 + \tfrac{1}{2}I\omega^2$$

Rolling without slipping: $v = r\omega$.

7 ROLLING DOWN A SLOPE FROM REST

$$mgh = \tfrac{1}{2}mv^2\left(1 + \frac{I}{mr^2}\right)$$

Body	v^2
Ring	gh
Disc	$\frac{4}{3}gh$
Sphere	$\frac{10}{7}gh$

Smaller I/mr^2 reaches the bottom first: sphere, then disc, then ring.

8 COMMON SECTION A TRAPS

- Using rev or degrees: convert to rad (1 rev = 2π rad).
- Forgetting the rotational KE of a rolling body.
- Using I about the centre when the axis is at the end.
- Claiming kinetic energy is conserved when the skater pulls in (it increases).

EXAM TIPS

✓Write $\tau = I\alpha$ for each rotating body and $F = ma$ for each hanging mass.

✓Link them with $a = r\alpha$.

✓Use angular momentum when the shape changes.

✓Use energy when heights are given.